RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF TECHNOLOGY

Bachelor of Information and Communication Technology
First Year - Semester I Examination – March/April 2025

ICT 1202 – ELECTRONIC CIRCUITS

Time: Two (02) hours

- Answer <u>**ALL**</u> Questions.
- Use of a non-programmable calculators is permitted.
- There are **FOUR (04)** essay questions printed on **THREE (03)** pages.

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1. a) Using the Kirchhoff's rules, determine the current through the **5V battery** in the network given in Figure 01.

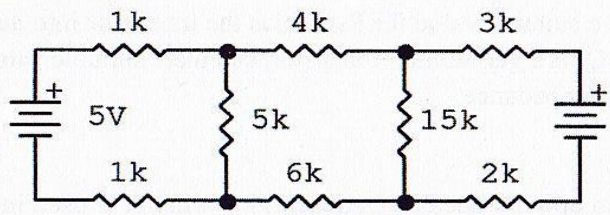

Figure 01

[10 Marks]

b) i. How does **p-type** and **n-type** doping increase the conductivity of a pure semiconductor?

ii. Why conductivity of a semiconductor increases with increasing temperature?

[10 Marks]

c) Illustrate the output signal produced by the clipping circuit in Figure 02, for a 6V (peak) sinusoidal input signal.

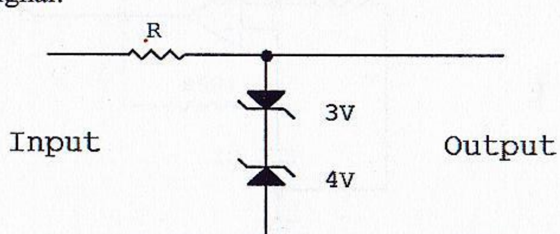

Figure 02

[05 Marks]
[Total 25 Marks]

2. a) Use the given circuit in Figure 03 to perform the following calculations.

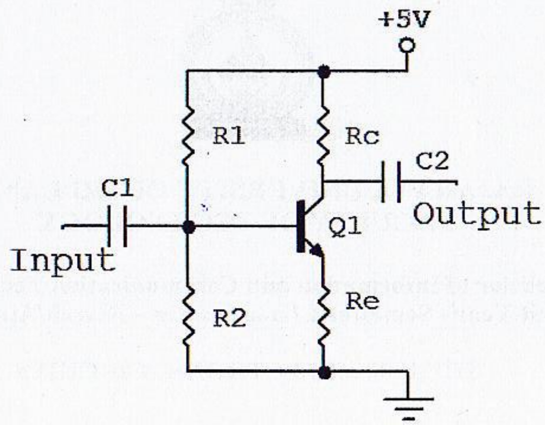

Figure 03

i. Choose a suitable collector voltage value (Vc) to design a **class A** amplifier for obtaining an output sinusoidal signal of 2 V peak.
ii. Calculate the resistance R_C and the base current I_B when the collector current is 2.5 mA and the current gain (β) is 100.
iii. Determine a suitable value for R_E to bias the transistor into **active region.**
iv. Assuming Q_1 is a germanium transistor, calculate suitable values for R_1 and R_2 with a low input impedance.

[Total 25 Marks]

3. a) Explain why an op-amp needs a feedback loop when it is used in circuits other than in a comparator circuit.

[05 Marks]

b) Explain the functioning of an op-amp circuit shown in Figure 04 (supply voltage of the Op-amp is 10 V and -10 V).

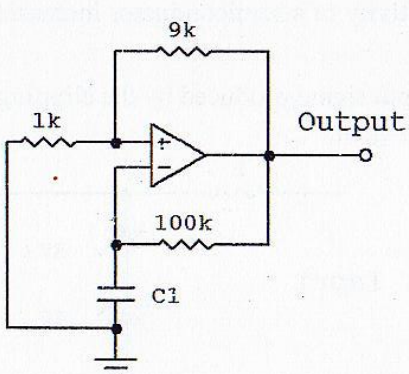

Figure 04

[10 Marks]

c) Using an ideal Op Amp model, calculate the output voltage for the circuit shown in Figure 05 for a power supply of +10 V and -10 V to the op-amp.

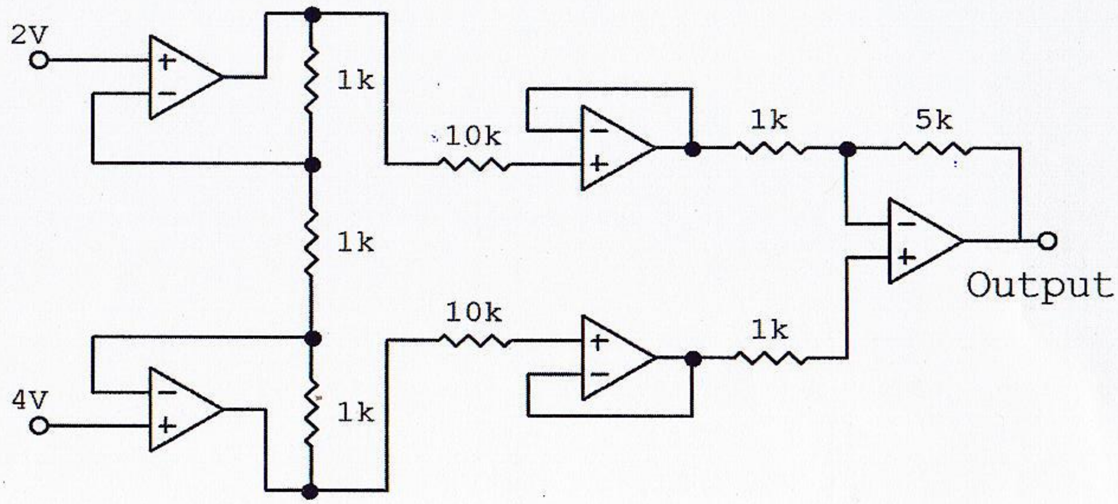

Figure 05

[10 Marks]
[Total 25 Marks]

4. The ***Fibonacci Sequence*** is a series of numbers. In this series, the next number is given by adding up the two numbers before it. Using this numbering system, the ***n***^th^ number of the *Fibonacci sequence* can be defined by the following three equations.

$$F_1 = 0$$
$$F_2 = 1$$
$$F_n = F_{n-1} + F_{n-2} \quad \text{for n} = 3, 4, 5, \ldots$$

A digital circuit takes four binary digits as an input and produces **1** as its output if the decimal value represented by the four binary digits is a ***Fibonacci number*** and otherwise produces **0**. Assume that all four binary digits represent positive decimal values (No bit is allocated for the sign).

a) Construct a truth table for this digital circuit.

[06 Marks]

b) Obtain the sum of products expression from the truth table.

[04 Marks]

c) Construct a K - Map and obtain a Boolean expression for the result (b).

[10 Marks]

d) Draw the logic circuit diagram for the Boolean expression obtained in part (c).

[05 Marks]
[Total 25 Marks]

--End--